ResearchGate

See discussions, stats, and author profiles for this publication at: https://www.researchgate.net/publication/226355968

Production of net-spinning caddisflies (Hydropsychidae) and black flies (Simuliidae) on rock outcrop substrate in a small southeastern Piedmont stream

Article *in* Hydrobiologia · May 1984

DOI: 10.1007/BF00007661

CITATIONS
34

READS
57

2 authors:

Mary C. Freeman
United States Geological Survey

148 PUBLICATIONS **7,677** CITATIONS

SEE PROFILE

J. Bruce Wallace
University of Georgia

221 PUBLICATIONS **21,229** CITATIONS

SEE PROFILE

All content following this page was uploaded by J. Bruce Wallace on 24 March 2016.

The user has requested enhancement of the downloaded file.

Production of net-spinning caddisflies (Hydropsychidae) and black flies (Simuliidae) on rock outcrop substrate in a small southeastern Piedmont stream

Mary C. Freeman & J. Bruce Wallace
Department of Entomology, University of Georgia, Athens, GA 30602, U.S.A.

Keywords: Hydropsychidae, Simuliidae, production, life histories, Piedmont stream

Abstract

Production by hydropsychids and simuliids inhabiting a Piedmont stream rock outcrop was estimated by the size–frequency method. Combined annual larval production was 10.0 g ash-free dry wt m^{-2}, 87% of which was contributed by a single caddisfly species, *Hydropsyche betteni* Ross. *Simulium tuberosum* (Lundström), *Symphitopsyche sparna* (Ross), and a species of *Cheumatopsyche* were abundant on the outcrop only during the summer. There was also a single fall cohort of *Simulium pictipes* Hagen. Turbulent flow and sediment scour apparently limited larval habitat availability on the outcrop during the winter and spring.

High production by *H. betteni* may be attributable in part to nutrient enrichment of the stream from agricultural runoff. Warm stream temperatures also contributed to production by enhancing larval growth rates. Prolonged recruitment and high growth rates in spring and summer resulted in overlapping cohorts in all species. Production:Biomass ratios ($P:\bar{B}$) for the hydropsychid spring–summer generations were 11–15, indicating that they developed through at least two generations during this period, and were thus annually trivoltine. The results of this study emphasize the importance of correcting size–frequency production estimates for larval development time, even when the species is collected throughout the sampling interval.

Introduction

Filter-feeders are an important component of energy-flow pathways in many streams and rivers. These animals collect and consume drifting material and thus slow the downstream loss of nutrients and energy from the stream system (Wallace *et al.*, 1977). Although their impact on total seston quantity may be negligible, filter-feeders can decrease seston quality by removing easily assimilated material, thereby influencing food availability to downstream communities (Maciolek & Tunzi, 1968; Carlsson *et al.*, 1977; McCullough *et al.*, 1979; Oswood, 1979; Benke & Wallace, 1980; Georgian & Wallace, 1981; Haefner & Wallace, 1981). Filter-feeder production, defined as tissue elaborated over time (Waters, 1977), represents nutrients and energy retained and thus available to predators in a given reach of stream.

The larvae of net-spinning caddisflies of the family Hydropsychidae (Trichoptera) and black flies (Diptera: Simuliidae) are widespread and often abundant stream filter-feeders. Considered together, species of these two families ingest a spectrum of particles ranging from drifting invertebrates to bacteria and colloidal material (Wallace & Merritt, 1980), and thus are potentially important faunal components in streams from headwaters to large rivers. The highest hydropsychid and simuliid larval production estimates have been reported for lake outflows, where dense filter-feeder populations are supported by lacustrine zooplankton, algae, and detritus and associated bacteria (Ulfstrand, 1968; Rhame & Stewart, 1976; Carlsson *et al.*, 1977;

Hydrobiologia 112, 3–15 (1984).
© Dr W. Junk Publishers, The Hague. Printed in the Netherlands.

Sheldon & Oswood, 1977; Oswood, 1979; Wotton, 1979; MacFarlane & Waters, 1982; Parker & Voshell, 1983). Parker & Voshell (1983) observed hydropsychid production in excess of 318 g dry wt m^{-2} a^{-1} below an impoundment on a Virginia Piedmont river, and Ulfstrand (1968) reported black fly production of 1 kg wet wt m^{-2} a^{-1} in a Swedish lake outlet.

Studies made in free-flowing stream systems indicate that filter-feeder production may also be substantial in larger streams and rivers. Nelson & Scott (1962) estimated that rock-outcrop filter-feeders in a Georgia Piedmont river produced at least 20.2 g ash-free dry wt (AFDW) m^{-2} a^{-1}, and Parker & Voshell (1983) reported net-spinning Trichoptera production of 9.4, 11.2 and 50.4 g dry wt m^{-2} a^{-1} at free-flowing sites in two Virginia piedmont rivers. In Southeastern Coastal Plain rivers, both hydropsychid and simuliid populations colonizing woody snag substrates have been estimated to produce from 10 to over 40 g AFDW m^{-2} snag surface a^{-1} (Benke *et al.*, 1979; Cudney & Wallace, 1980).

Hydropsychid and simuliid production values available for small streams are low, mostly less than 1 g AFDW m^{-2} a^{-1} for any single species (Cushman *et al.*, 1977; Wotton, 1978; Neves, 1979; Benke & Wallace, 1980; Ross & Wallace, 1983). However, Haefner & Wallace (1981) reported that the hydropsychid production in a nutrient-enriched first-order Appalachian stream was almost 5 g AFDW m^{-2} a^{-1}, over 3 times that in a comparable but unenriched stream. Similarly, Hopkins (1976) estimated that *Hydropsyche* sp. produced 15 g dry wt m^{-2} a^{-1} in a New Zealand stream draining farmland.

The purpose of this study was to estimate filter-feeder production in a small southeastern Piedmont stream enriched by agricultural runoff, where it was anticipated that production could approach that observed in much larger streams.

Study site

This research was conducted on a granite outcrop in Rose Creek, Oconee County, Georgia, a tributary to the Oconee River in the Altamaha River drainage system. At the study site, Rose Creek is a third order stream and drains 1560 ha; approximately 62% of this area is used for pasture, poultry production, or grain cultivation. There is also some residential development in the catchment. Wooded land is concentrated around the branches of Rose Creek and consists of mixed pine and hardwood.

The outcrop is approximately 50 m long and varies from 10 to 20 m in width. The stream is shallow and characterized by rapid turbulent flow, falling approximately 7 m over the length of the outcrop. Stream width varies from about 4 to 10 m, depending on discharge. The outcrop is open to sunlight, and during periods of low discharge, periphyton is abundant in areas covered by the stream. There is notably little macrophyte growth on the outcrop, whereas such substrates are often covered with thick growths of *Podostemum ceratophyllum* in other Piedmont streams (Nelson & Scott, 1962; Caldwell, 1973; pers. observ.).

Above the outcrop, the stream substrate is predominantly sand and deposited silt and offers much less suitable habitat for hydropsychid and simuliid larvae. There is a stretch of boulder outcroppings for about 75 m immediately upstream from the outcrop and dense clusters of hydropsychids were observed on these rock surfaces.

Materials and methods

Sampling and sorting procedures

The outcrop benthos was sampled from October 1979 through October 1980. Samples were collected monthly from November through March, and biweekly from April through October. On each sampling date, eight Surber samples (0.3 m × 0.3 m, net mesh = 223 μm) were taken at random locations on the outcrop. Pools and sites where the surface was too uneven for the sampler to fit tightly against the rock were excluded from sampling, as were areas where the stream border fluctuates widely over the outcrop in response to changes in discharge. Foam weatherstripping was attached around the bottom edge of the sampler to avoid leakage between the sampler and the rock. Detritus and animals within the sampler area were scrubbed from the rock with a stiff brush. Samples were preserved in 5% formalin with phloxine B added, and returned to the laboratory for sorting. Because the stream is too shallow on the outcrop to permit the use of most current meters, current velocity and stream depth

were measured at the location of each sample with an aluminum velocity head rod (Wilm & Storey, 1944).

Animals were hand-picked from the samples under a dissecting microscope (12×). The hydropsychid larvae in each sample were identified, sorted by instar and counted. Larval instars were separated by head capsule width at eye level, which was measured with an ocular micrometer to the nearest 20 μm. Setal characters given by Mackay (1978) facilitated the identification of early instars. Simuliid larvae were identified and sorted into size classes based on head capsule post-genal length, also measured with an ocular micrometer to the nearest 20 μm. Instars could not easily be separated except for the first instars, which have an egg-burster, and final instars, distinguished by the presence of the pupal respiratory histoblast. Since the size-frequency production method assumes that all individuals can potentially grow to the same maximum size (Hamilton, 1969), the final instars were separated as one size class (size class 7) which included a large range of post-genal lengths. Larvae falling between the first and last instars were separated into equal size classes (Table 1).

Larval weights for each instar (size class) were estimated by weighing preserved larvae on a Cahn electrobalance to the nearest μg. Larvae were individually dried on foil pans at 65 °C for 24 h, desiccated for 24 h, and weighed. They were then ashed (485 °C) for 1 h, cooled in a desiccator (24 h), and reweighed to estimate larval ash-free dry weight (AFDW).

The coarse particulate organic matter (CPOM: leaves, and leaf and stick fragments longer than approximately 0.5 cm), and the remaining detritus and sediment in each sample were dried separately at 65 °C, desiccated for at least 24 h, and weighed. The two fractions were then ashed at 485 °C, cooled in a desiccator, and reweighed to obtain the ash-free dry weight of CPOM and the smaller detritus.

Production calculations

Annual production of each hydropsychid and simuliid species was estimated by the size-frequency method (Hynes & Coleman, 1968; Hamilton, 1969; Waters & Hokenstrom, 1980). As suggested by Krueger & Martin (1980), the mean annual number of individuals in each instar (size class) was weighted for sampling interval length, and the larval weight at loss was estimated by the geometric mean of the weights of individuals in successive instars (size classes). The production estimates were corrected for the average length of time required for development from hatching to pupation, or the cohort production interval (CPI), as described by Benke (1979). The species CPIs were estimated from instar frequency patterns through the year. The computer program developed by Krueger & Martin (1980) was used to calculate 95% confidence intervals for the annual production estimates.

Production was also estimated by the removal-summation method (Waters & Crawford, 1973) for those species in which cohort development was sufficiently synchronous so that both mean larval weight and mean population density were correlated with time.

Table 1. Simuliid larval size classes, based on post-genal length, and mean ash-free dry weights used in production calculations. Standard errors of all weight estimates are less than 10% of the mean, except as indicated. Number of individuals weighed is shown in parentheses.

	Simulium pictipes		*Simulium tuberosum*	
Size class	Post-genal length (μm)	Larval AFDW (mg)	Post-genal length (μm)	Larval AFDW (mg)
1	<120	0.002 (6)[a]	< 80	0.001 (2)[b]
2	140–200	0.006 (15)[a]	100–140	0.006 (25)
3	220–280	0.030 (31)	160–200	0.012 (28)
4	300–360	0.083 (29)	220–260	0.038 (26)
5	380–440	0.266 (30)	280–320	0.075 (47)
6	460–520	0.486 (38)	>340	0.217 (47)
7	>540	1.486 (51)	–	–

[a] Standard error is 15 to 20% of the mean.
[b] Standard error was not calculated.

Results and discussion

Species collected

Three species of hydropsychids inhabited the outcrop. *Hydropsyche betteni* Ross, a species widely distributed in Eastern North America and noted for its tolerance to organic pollution (Schuster & Etnier, 1978), was the most abundant caddisfly. *Symphitopsyche sparna* (Ross), a widespread species which commonly occurs with *H. betteni* (Schuster & Etnier, 1978), and a species of *Cheumatopsyche* were also collected.

Simulium pictipes Hagen and *Simulium tuberosum* (Lundström) were the most abundant simuliid species at the study site. Both species are widely distributed in temperate streams. *Simulium pictipes* larvae commonly aggregate on bedrock substrates in rapids and waterfalls, whereas *S. tuberosum* larvae occur in a variety of habitats, including impoundment outflows and areas in high current velocities (Stone & Snoddy, 1969; Kurtak, 1979). In addition, isolated *Simulium jonesi* Stone and Snoddy and *Simulium notiale* Stone and Snoddy pupae were collected in the fall, and it is possible that some larvae of other species were included with *S. tuberosum*.

Larval abundance and growth

Filter-feeder densities were highest on the outcrop from late spring to early fall. *Hydropsyche betteni* was the only hydropsychid species collected throughout the year, and samples taken from November 1979 to April 1980 averaged less than 170 *H. betteni* larvae m^{-2}. The *H. betteni* population density increased over 10-fold from May to late July 1980, peaking at approximately 3 000 larvae m^{-2}, and then declined to less than 600 larvae m^{-2} by September and October. This trend was repeated by *S. sparna* and *Cheumatopsyche* sp., which were abundant only during June, July and August, and by *S. tuberosum*, abundant from May to August. Only isolated individuals of these three species were collected during the remainder of the year. *Simulium pictipes*, the only exception to this trend, was represented by high population densities in October 1979 (> 1400 larvae m^{-2}), but the population declined sharply in November and through the winter and was rarely collected in 1980.

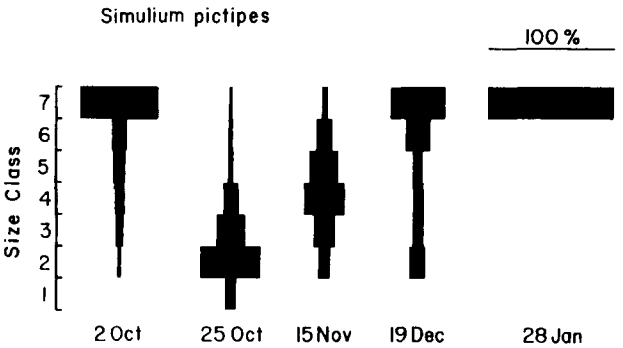

Fig. 1. Simulium pictipes larval population structure in Rose Creek from 2 October 1979 to 28 January 1980. Only one first instar larva was collected on 15 November and only one larva was collected on 28 January. The 2 October through 28 January samples all contain at least 90 larvae.

The pattern of size-class frequencies for *S. pictipes* (Fig. 1) shows that the 2 October samples were comprised primarily of final instars and that a cohort subsequently developed from late October through December or January. In contrast, the hydropsychid and *S. tuberosum* larvae developed rather asynchronously (Figs. 2–5), confounding the estimation of cohort production intervals. For example, although the *H. betteni* larvae may be roughly divided into spring-summer vs. winter generations (Fig. 2), the continuous recruitment of first instars from April through November makes it difficult to estimate the time actually required for an individual to complete development.

There was an apparent spring generation of *S. sparna* from 21 April to 26 June (66 days), and, in addition, a fairly well defined summer cohort developed from 26 June to 22 August (57 days) (Fig. 3). Based on these *S. sparna* cohorts, we conservatively estimated the hydropsychid spring and summer CPIs as 66 days. Mackay (1979) observed similar growth rates for *H. betteni* and *S. sparna* in two southern Ontario rivers, where larval recruitment began in late May and early June, followed by rapid larval growth to instar V by early July and pupation in August. The *H. betteni* winter CPI was estimated as 161 days, corresponding to the interval from early October to mid March.

It is difficult to distinguish individual generations of *S. tuberosum* from the pattern of size class frequencies (Fig. 5). Stone & Snoddy (1969) reported that this species has at least 5 generations annually

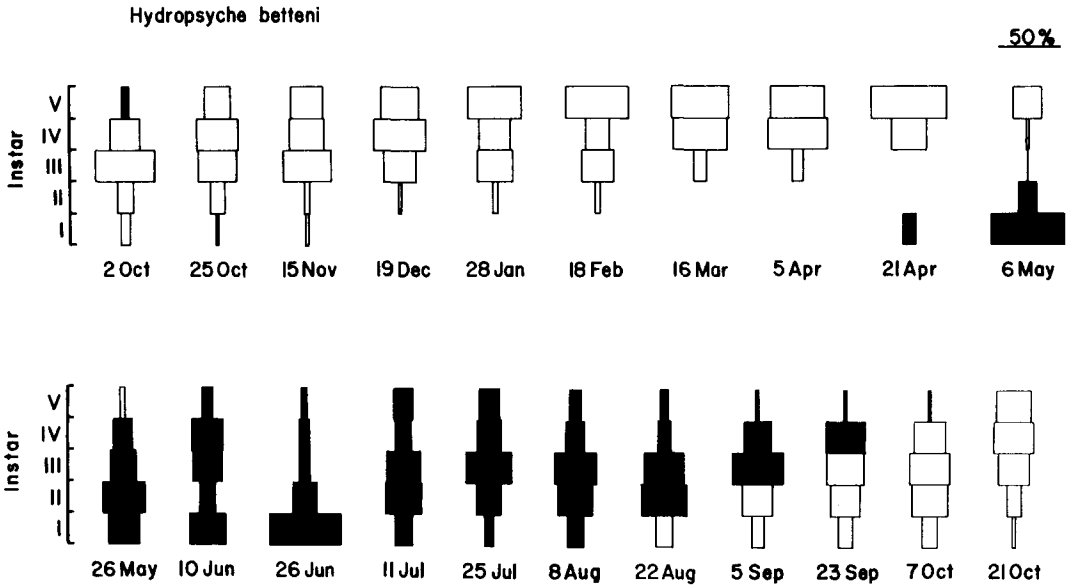

Fig. 2. Hydropsyche betteni larval population structure in Rose Creek, October 1979–October 1980. Darkened bars represent larvae assigned to the spring and summer generations; open rectangles represent winter generation larvae. Only samples collected on 16 March and 21 April contain fewer than 50 individuals.

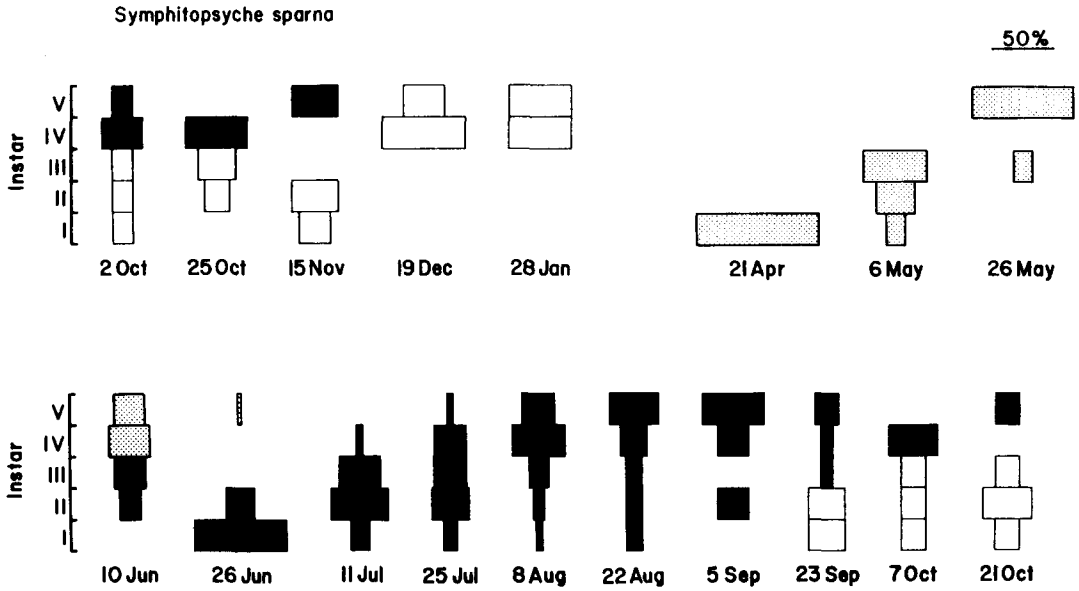

Fig. 3. Symphitopsyche sparna larval population structure in Rose Creek, October 1979–October 1980. Shaded bars represent spring generation larvae and darkened bars represent larvae assigned to the summer generation. Open rectangles represent winter larvae which were excluded from production calculations. Only samples collected from 26 June through 8 August contain over 50 individuals. No larvae were collected on 18 February, 16 March, or 5 April. Only one larva was collected on 21 April.

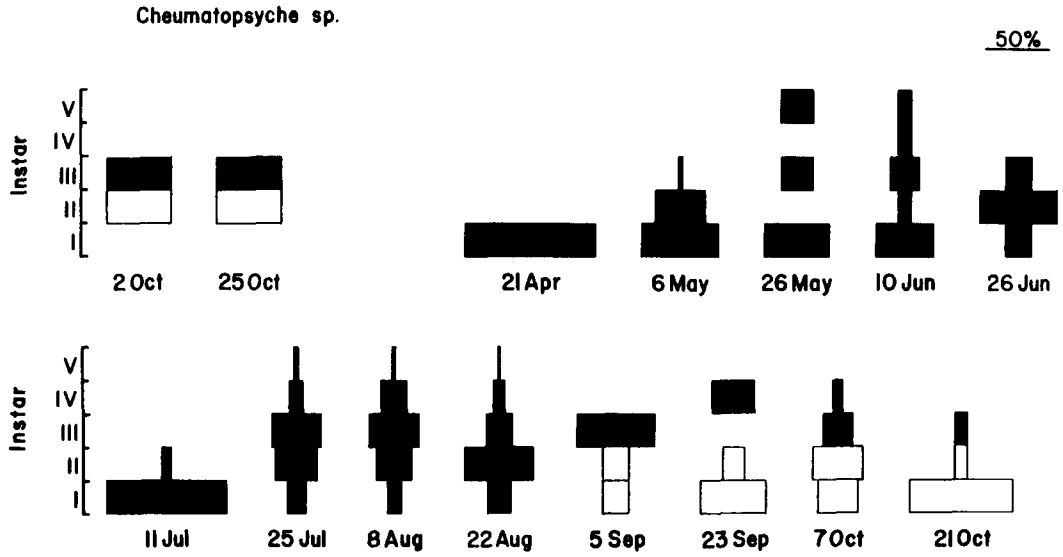

Fig. 4. Cheumatopsyche sp. larval population structure in Rose Creek, October 1979–October 1980. Darkened bars represent larvae assigned to the spring and summer generations; open rectangles represent winter larvae which were excluded from production calculations. Only samples collected from 25 July through 22 August contain more than 50 individuals. No larvae were collected on 19 December, 16 March, or 5 April. Only one larva was collected on 15 November, 28 January, and on 18 February.

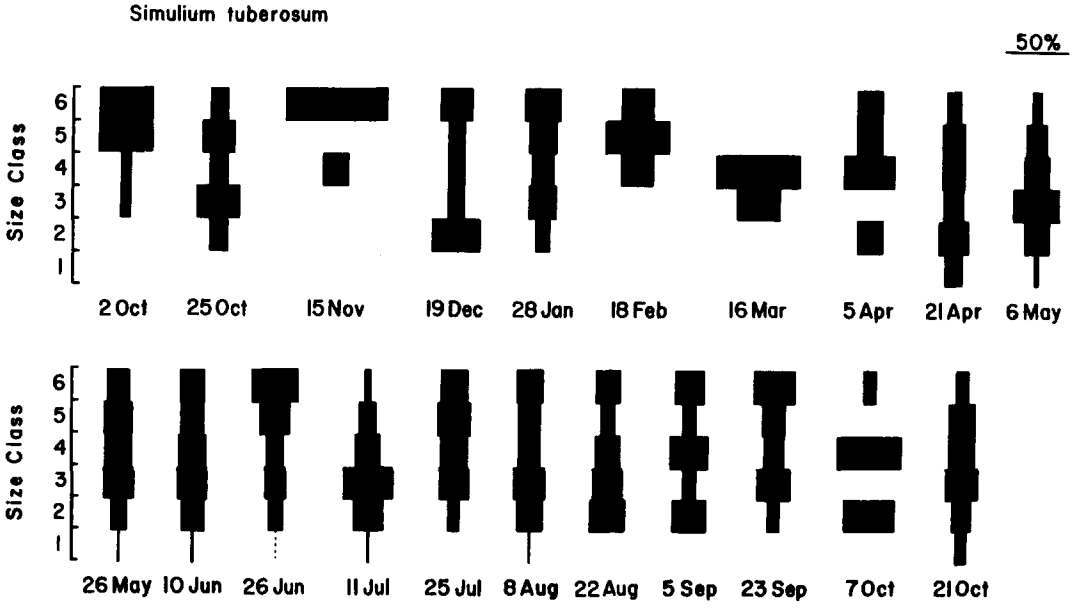

Fig. 5. Simulium tuberosum larval population structure in Rose Creek, October 1979–October 1980. On 26 June, only two first instar larvae were collected. Samples collected from 6 May through 8 August, and on 21 October and 25 October contain a minimum of 50 individuals.

Table 2. Mean larval ash-free dry weights (mg), ± standard errors, used in size-frequency production calculations. The total number of larvae weighed is shown in parentheses. The annual weights (A) shown for *H. betteni* are averages of monthly means, with the number of monthly values used shown in parentheses. In the seasonal calculations, the summer and winter weights were pooled for *H. betteni* first and second instar larvae.

Species	Instar				
	I	II	III	IV	V
Cheumatopsyche sp.	0.005 ± 0.0005 (23)	0.019 ± 0.001 (28)	0.043 ± 0.003 (37)	0.133 ± 0.009 (45)	0.674 ± 0.066 (39)
Symphytopsyche sparna	0.008 ± 0.0006 (25)	0.022 ± 0.002 (42)	0.081 ± 0.005 (36)	0.285 ± 0.021 (46)	1.178 ± 0.153 (45)
Hydropsyche betteni					
(A)	0.011 ± 0.0003 (2)	0.045 ± 0.007 (3)	0.241 ± 0.033 (4)	1.025 ± 0.166 (6)	5.120 ± 1.195 (6)
Summer	0.011 ± 0.001 (47)	0.045 ± 0.003 (51)	0.192 ± 0.016 (37)	0.726 ± 0.045 (62)	3.015 ± 0.127 (168)
Winter	0.011 ± 0.001 (47)	0.045 ± 0.003 (51)	0.275 ± 0.014 (34)	1.324 ± 0.078 (60)	7.452 ± 0.328 (172)

Table 3. Size-frequency production calculations for the spring and summer generations (21 April-7 October, 169 days) and winter generation (22 August-6 May, 258 days) of *Hydropsyche betteni*. The CPI estimates are 66 days for the spring and summer, and 161 days for the winter.

Instar	Mean density (no. m $^{-2}$)	Mean ind. wt. (mg)	Stand. stock (mg m $^{-2}$)	No. lost (no. m $^{-2}$)	Wt at loss (mg)	Wt lost × 5 (mg m $^{-2}$)
Spring and summer generations						
I	190.824	0.011	2.099			
				-32.378	0.022	-3.560
II	223.202	0.045	10.044			
				-57.932	0.093	-26.938
III	281.134	0.192	53.978			
				106.359	0.373	198.360
IV	174.775	0.726	126.887			
				69.073	1.479	510.795
V	105.702	3.015	318.693			
			B = 511.70	105.702	3.015	1593.458
						Sum = 2272.11
P = 2272.1 × 169/66 = 5818.0 mg m $^{-2}$; P:$\bar{B}$ = 11.4.						
Winter generation						
I	16.856	0.011	0.185			
				-18.515	0.022	-2.035
II	35.371	0.045	1.592			
				-34.907	0.111	-19.375
III	70.278	0.275	19.326			
				7.039	0.603	21.225
IV	63.239	1.324	83.728			
				-7.222	3.141	-113.420
V	70.461	7.452	525.075			
			B = 629.91	70.461	7.452	2625.375
						Sum = 2511.77
P = 2511.8 × 258/161 = 4025.1 mg m $^{-2}$; P:$\bar{B}$ = 6.4						

in Alabama streams. We conservatively estimated the CPI as 44 days, based on shifts in larval abundance from small to large size classes over the following intervals: early October to mid November (46 days); mid December to mid February (64 days); mid-April to early June (50 days); 1 July to 1 August (31 days); and mid August to mid September (31 days).

Production and Turnover rates

Hydropsyche betteni production comprised almost 98% of the total hydropsychid production on the outcrop. Production was similar between the summer and winter generations, 5818 vs. 4025 mg AFDW m^{-2} respectively (Table 3), despite depressed winter population densities. However, the average mass of winter fifth instars was approximately 2.5 times that of larvae maturing in the summer (Table 2, Fig. 6). The sum of summer and winter production estimates was within the 95% confidence interval of annual production calculated using mean larval weights and a mean CPI of 114 days (Table 4). However, we agree with reservations expressed by Hynes (1980) and MacFarlane & Waters (1982) that estimates of the 95% CI (25.7 and 31.3% of production for *H. betteni* and *S. tuberosum*, respectively, Table 4) using the Krueger & Martin (1980) program seem surprisingly low when compared with the high variance inherent with stream benthic sampling. The annual P:$\bar{B}$ ratio (production:mean biomass) was 13.7, higher than the value of approximately 10 predicted for bivoltine species (Waters, 1977).

Production was calculated for the spring and summer generations only of *S. sparna* and *Cheumatopsyche* sp. by excluding those larvae belonging to the winter generation, as indicated in Figs. 3 and 4. These estimates, corrected for a CPI of 66 days, were 185.4 mg AFDW m^{-2} for *S. sparna* and 130.7 mg AFDW m^{-2} for *Cheumatopsyche* sp. The P:$\bar{B}$ values were 14.3 and 14.7, respectively, indicating that the hydropsychids developed through at least two generations during this period, and possibly had a partial third generation in the late summer. The high summer (11.4) and annual P:$\bar{B}$ values obtained for *H. betteni* likewise indicate that the population was trivoltine, developing through winter, spring and summer generations.

The interpretation of high turnover ratios as indicators of polyvoltinism admittedly involves circular reasoning given that the production estimates have already been adjusted for short larval development times via the CPI correction. In the case of

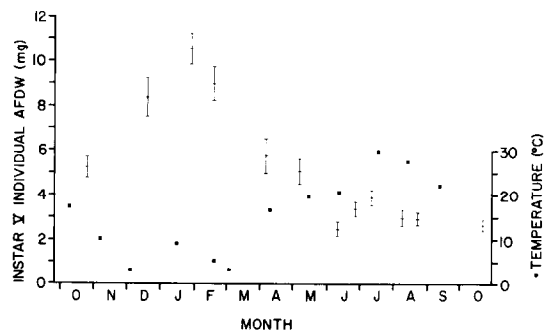

Fig. 6. Mean ash-free dry weights, ± standard errors, of *Hydropsyche betteni* fifth instar larvae on selected dates from October 1979 through October 1980. Temperature readings made by the USGS in the Oconee River near Rose Creek are also indicated (■).

Table 4. Summary of filter-feeder production parameters in Rose Creek, 1979–1980. Annual means are shown for *H. betteni* and *S. tuberosum*, and 95% confidence intervals are given for the production estimates of these species. The values shown for *S. sparna* and *Cheumatopsyche* sp. are for the spring and summer generations only. Similarly, only the fall cohort of *S. pictipes* is included.

Species	CPI (days)	N (no. m^{-2})	B (mg m^{-2})	P (mg m^{-2})	P:$\bar{B}$
Hydropsyche betteni	114	606.0	637.8	8761.5 (6509–11014)	13.7
Symphitopsyche sparna	66	75.3	13.0	185.4	14.3
Cheumatopsyche sp.	66	139.9	8.9	130.7	14.7
Simulium tuberosum	44	326.2	23.6	711.7 (489– 937)	30.2
Simulium pictipes	–	352.9	61.4	337.3	5.5

S. sparna, however, the spring and summer cohorts developed synchronously enough for production to be estimated by the removal-summation method (Table 5), which does not require any estimate of larval development time. Thus calculated, the spring cohort produced 51.4 mg AFDW m^{-2}, whereas summer production was 94.7 mg AFDW m^{-2}. Note that although spring population densities were low in comparison with summer densities, *S. sparna* spring production was relatively high. This results from the larger size of the spring V instars, mean wt = 3.669 mg on 26 May (n = 5), as compared to a mean wt of 0.849 mg for V instars collected from July through August. The P:$\bar{B}$ values derived from the removal-summation estimates were 4.9 and 7.3 for the spring and summer generations, respective-

Table 5. Removal-summation production calculations for the spring and summer generations of *Symphitopsyche sparna*.

Date	no. m^{-2}	mg m^{-2}	$\bar{x}$ ind. wt (mg)	No. lost	Wt at loss	Wt lost mg m^{-2}
Spring generation, 21 April–26 June						
21 Apr	1.346	.011	.008			
				-24.219	.030	-.727
6 May	25.565	1.302	.051			
				17.492	1.561	27.305
26 May	8.073	24.792	3.071			
				1.346	2.403	-3.234
10 June	9.419	16.345	1.735			
				6.728	2.702	18.179
26 June	2.691	9.873	3.669			
				2.691	3.669	9.873
		B = 10.5 mg m^{-2}				P = 51.4 mg m^{-2}
			P:$\bar{B}$ = 4.9			
Summer generation, 10 June–21 Oct. + 15 Nov.						
10 Jun	6.728	.386	.057			
				-222.007	.034	-7.548
26 Jun	228.735	2.602	.011			
				1.345	.032	0.043
11 Jul	227.390	12.143	.053			
				-157.423	.101	-15.900
25 Jul	384.813	57.108	.148			
				254.299	.256	65.101
8 Aug	130.514	47.389	.363			
				99.567	.386	38.433
22 Aug	30.947	12.647	.409			
				25.565	.455	11.632
5 Sept	5.382	2.698	.501			
				0	.509	0
23 Sept	5.382	2.777	.516			
				2.691	.401	1.079
7 Oct	2.691	.767	.285			
				1.345	.567	.763
21 Oct	1.346	1.143	.849			
				-2.691	.849	-2.285
15 Nov	4.037	3.427	.849			
				4.037	.849	3.427
		B = 13.0 mg m^{-2}				P = 94.7 mg m^{-2}
			P:$\bar{B}$ = 7.3			

ly. Single cohorts of other stream benthos have been shown to have P:$\bar{B}$ values ranging from 4 to 6 (Waters, 1977).

These calculations again indicate that the hydropsychids had both a spring and a summer generation, and that there may have been a partial second summer generation. Correcting the annual size-frequency production values with a CPI of 6 months, the interval during which the summer larvae were available for sampling, would thus underestimate summer production by over 60 percent.

The total hydropsychid production, primarily by *H. betteni,* was thus 9 g AFDW m^{-2} a^{-1}: higher than observed in other small streams and similar to lower estimates from Piedmont and Coastal Plain rivers. Gut content analyses have demonstrated the importance of carnivory and diatom ingestion in many hydropsychid species (Coffman *et al.,* 1971; Benke & Wallace, 1980; Fuller & Mackay, 1980; Ross & Wallace, 1983), suggesting that the low hydropsychid production in headwater streams results from the limited availability of high quality food (Georgian & Wallace, 1981). In Rose Creek, enrichment by agricultural runoff may contribute to hydropsychid productivity by subsidizing algal growth and production of potential prey species. However, production is also enhanced as a result of the mild winter and elevated summer stream temperatures which allow the hydropsychids to develop through 3 generations a year. Monthly temperature readings made by the USGS in the Oconee River near Rose Creek during 1979 and 1980 (USGS, 1980) indicate accumulations of approximately 2000 degree-days from September to April and from April to June, and of approximately 2400 degree-days from June to September. Thus, larvae hatching in early spring may have experienced as many degree-days as the entire winter generation by mid June, and presumably complete development.

Variability in growth rate and voltinism as a function of temperature regimes has been observed for other aquatic insects (Ward & Stanford, 1982). Similarly, the flexibility of hydropsychid development time is becoming apparent as species life histories are studied in streams with differing thermal regimes and nutrient inputs. Univoltine and bivoltine populations of *H. betteni* and *S. sparna* have been observed in two southern Ontario rivers, bivoltinism occurring at the more downstream sites (Mackay, 1979). Mackay (1979) also saw evidence of a partial third generation of *S. sparna* at one downstream site. Generally, headwater hydropsychid populations appear to be predominantly univoltine (e.g. Benke & Wallace, 1980; Ross & Wallace, 1983), whereas bivoltinism is more common in larger rivers (Benke *et al.*, 1979; Cudney & Wallace, 1980). The only other reported observations of trivoltine hydropsychid populations are from two Virginia Piedmont rivers (Parker & Voshell, 1982).

One result of the rapid spring and summer hydropsychid development in Rose Creek is the smaller mass of the summer final instars as compared to larvae maturing in the winter (Table 2, Fig. 6). Vannote & Sweeney (1980) predicted that in polyvoltine aquatic insects, adult size should vary between generations as a result of the temperature dependency of development rates. They also hypothesized that the distributions of some aquatic species are limited by diminished body size and therefore reduced fecundity resulting from development at non-optimal temperatures (Vannote & Sweeney, 1980). In the case of *H. betteni* and *S. sparna*, the ability to shift from univoltinism to polyvoltinism depending on the local thermal regime, even with the accompanying reduction in size at maturity, apparently allows these species to occur over a broad geographical range.

The outcrop simuliids also developed rapidly, but sparse population densities resulted in low overall production: 712 mg AFDW m^{-2} a^{-1} by *S. tuberosum* and 337 mg AFDW m^{-2} by the single fall cohort of *S. pictipes* (the latter calculated by the removal-summation method). Annual *S. tuberosum* production was similar to that reported for Michigan stream simuliids (Merritt *et al.*, 1982) and approximately 2 times simuliid production in both a temperate woodland stream (Neves, 1979) and a southern Appalachian stream (J. B. Wallace & D. H. Ross, unpubl. data). The outcrop simuliid production was much lower than estimates from coastal plain river snags (Benke *et al.*, 1979), and was also low compared with the production of 6.1 g dry wt m^{-2} a^{-1} reported by Ladle *et al.* (1972) for simuliids inhabiting macrophytes in an English chalk stream. This study does indicate, however, the high production potential of *S. pictipes* in areas where dense aggregations of these large larvae are present throughout the year.

Limiting factors

Population densities of simuliids and hydropsychids were possibly limited throughout the year, and especially in the winter and spring, by unfavorable flow conditions and sediment scour on the outcrop. Mean current velocity at the sample locations ranged from 75 to 106 cm s^{-1} during the summer, and was higher during the remainder of the year (Table 6). Mean depth ranged from 1.2 to 3.5 cm. This rapid, turbulent flow on some dates carried a considerable sediment load. The high sample ash-weight values in February, April, and May are the results of sand washing over the outcrop and collecting in the Surber net during sampling. As discharge decreased from late spring into summer, corresponding decreases in current velocity and suspended sand may have made more habitat available for the filter-feeders. Alternatively, because the stream was narrower at lower discharges, the summer increases in larval densities may be attributable to concentration of the larvae in a smaller total stream area. Although the larvae probably did become concentrated during mid-summer extreme low flow conditions, summer population increases over the winter levels were too large to be explained by shrinking stream area alone.

Fowler's Mill Creek, another tributary to the Oconee River, is of similar size to Rose Creek and also flows over a granite outcrop. Caldwell (1973) estimated that this outcrop, which has a lower gradient than the Rose Creek study site and is covered by thick mats of *Podostemum ceratophyllum*, supported a mean *Hydropsyche* spp. population density of over 2900 larvae m^{-2} a^{-1}. This is more than four times the annual hydropsychid density observed in Rose Creek. Moreover, the Fowler's Mill hydropsychid densities peaked in November and were high through most of the winter and spring (Caldwell, 1973).

The absence of macrophyte growth may also be indicative of harsh conditions at the study site. In their study on an outcrop in the Middle Oconee River, Nelson & Scott (1962) saw that high turbulent flow resulted in a loss of *Podostemum* from the study area and reduced faunal production. *Podostemum* may similarly be limited on the Rose Creek

Table 6. Mean current velocity and depth at the sample locations, and mean weights of ash and of coarse (>0.5 cm) and smaller detritus (<0.5 cm) in the samples collected in Rose Creek, 1979–1980. Standard errors are shown in parentheses; eight samples were collected on each date.

Date	Current velocity (cm s^{-1})	Depth (cm)	Detrital AFDW		Ash wt (g m^{-2})
			>0.5 cm (g m^{-2})	<0.5 cm (g m^{-2})	
15 Nov.	112 (14)	2.3 (.67)	4.3 (1.1)	1.4 (0.2)	35.0 (12.1)
19 Dec.	115 (14)	1.5 (.39)	1.1 (0.3)	3.4 (0.6)	216.6 (97.9)
28 Jan.	118 (09)	1.2 (.41)	1.0 (0.3)	4.1 (1.5)	297.8 (76.9)
18 Feb.	142 (11)	2.3 (.80)	2.4 (0.6)	8.3 (1.5)	897.0 (297.6)
16 Mar.	133 (12)	2.4 (.65)	1.3 (0.5)	4.2 (1.1)	324.8 (126.9)
05 Apr.	134 (08)	2.9 (1.15)	4.5 (2.5)	9.8 (1.8)	1694.2 (532.8)
21 Apr.	117 (08)	3.5 (.89)	6.6 (2.3)	14.0 (3.5)	2187.8 (854.9)
06 May	103 (11)	2.4 (.63)	2.6 (1.2)	12.8 (5.1)	1273.1 (614.0)
26 May	129 (09)	1.4 (.25)	0.5 (0.2)	1.9 (0.4)	89.0 (57.2)
10 Jun.	105 (14)	1.6 (.32)	1.6 (0.4)	4.2 (0.9)	458.4 (273.1)
26 Jun.	89 (11)	2.2 (.91)	1.3 (0.5)	3.7 (0.9)	464.2 (150.5)
11 Jul.	95 (15)	2.6 (.99)	2.0 (0.8)	6.7 (2.7)	685.5 (358.7)
25 Jul.	75 (09)	1.7 (.46)	30.5 (24.5)	8.4 (1.8)	385.1 (71.9)
08 Aug.	106 (18)	2.7 (.46)	7.2 (6.1)	5.4 (1.2)	569.4 (149.4)
22 Aug.	89 (15)	2.1 (.65)	1.8 (0.4)	3.0 (0.6)	229.7 (53.6)
05 Sept.	100 (10)	3.3 (1.42)	2.6 (0.5)	4.9 (1.2)	693.5 (216.4)
23 Sept.	91 (12)	0.8 (.13)	1.3 (0.9)	1.4 (0.2)	92.7 (29.2)
07 Oct.	90 (12)	1.3 (.70)	6.7 (5.2)	2.1 (0.6)	167.2 (64.8)
21 Oct.	100 (09)	3.4 (1.83)	8.3 (2.5)	2.3 (0.5)	155.9 (56.6)

outcrop by turbulent flow combined with sediment scour. To the extent that macrophytes provide attachment sites and refugia from the current (Williams & Hynes, 1973; Cudney & Wallace, 1980), the lack of *Podostemum* further limits the outcrop hydropsychid densities.

Summary and conclusions

Rose Creek filter-feeder production was lower than that reported for hydropsychids by Parker & Voshell (1983) and for filter-feeders by Nelson & Scott (1962) in larger Piedmont rivers. However, the value reported here is probably a minimal estimate of the production potential in Piedmont streams, where adequate substrate is available, because harsh flow conditions limited population densities at the study site for much of the year. Despite this limitation, the *H. betteni* production in Rose Creek was eight times higher than the combined production of 6 caddisfly species in the 4th-order Tallulah River, located in the North Georgia mountains (Benke & Wallace, 1980).

Stream temperature has been interpreted as the primary factor controlling invertebrate growth rates because it may influence both metabolism and food abundance and quality (Anderson & Cummins, 1979). Similarly, warmer temperatures in Piedmont and upland rivers as compared with those at higher elevations may contribute to hydropsychid productivity by enhancing prey production and shortening larval development times. It is interesting that the only reports of trivoltine hydropsychid populations are from Piedmont streams (i.e. this study, Parker & Voshell, 1982). In this study, the P:$\bar{B}$ of 7.3 for *S. sparna* from June to November (as calculated by the removal-summation method) suggests the possibility of a partial 4th generation developing in the late summer and early fall. This is not unreasonable considering that, in addition to a summer generation, a partial generation of *S. sparna* has also been reported for late summer in cooler streams (Mackay, 1979; Ross & Wallace, 1983).

Acknowledgements

The authors thank S. Martin, J. O'Hop, and D. Ross for assistance. Drs. C. W. Berisford, D. A. Crossley, Jr., and P. E. Hunter provided valuable criticism on an earlier draft of the manuscript.

References

Anderson, N. H. & K. W. Cummins, 1979. Influences of diet on the life histories of aquatic insects. J. Fish. Res. Bd Can. 36: 335–342.

Benke, A. C., 1979. A modification of the Hynes method for estimating secondary production with particular significance for multivoltine populations. Limnol. Oceanogr. 24: 168–171.

Benke, A. C., D. M. Gillespie, F. K. Parrish, T. C. Van Arsdall, R. J. Hunter & R. L. Henry III, 1979. Biological basis for assessing impacts of channel modification: invertebrate production, drift, and fish feeding in a southeastern blackwater river. Envir. Res. Cent. Ga. Inst. Tech. Atlanta Rep. 06–79, 187 pp.

Benke, A. C. & J. B. Wallace, 1980. Trophic basis of production among net-spinning caddisflies in a southern Appalachian stream. Ecology 61: 108–118.

Caldwell, S. D., 1973. Ecology of the aquatic insects in Fowler's Mill Creek. Ph.D. Diss., Univ. Georgia, Athens, Ga., 166 pp.

Carlsson, M., L. M. Nilsson, Bj. Svensson, S. Ulfstrand & R. S. Wotton, 1977. Lacustrine seston and other factors influencing the blackflies (Diptera: Simuliidae) inhabiting lake outlets in Swedish Lapland. Oikos 29: 229–238.

Coffman, W. B., K. W. Cummins & J. C. Wuycheck, 1971. Energy flow in a woodland stream ecosystem, 1. Tissue support trophic structure of the autumnal community. Arch. Hydrobiol. 68: 232–276.

Cudney, M. D. & J. B. Wallace, 1980. Life cycles, microdistribution and production dynamics of six species of net-spinning caddisflies in a large southeastern (U.S.A.) river. Holarct. Ecol. 3: 169–182.

Cushman, R. M., J. W. Elwood & S. G. Hildebrand, 1977. Life history and production dynamics of Alloperla mediana and Diplectrona modesta in Walker Branch, Tennessee. Am. Midl. Nat. 98: 354–364.

Fuller, R. L. & R. J. Mackay, 1980. Feeding ecology of three species of Hydropsyche (Trichoptera: Hydropsychidae) in southern Ontario. Can. J. Zool. 58: 2239–2251.

Georgian, T. J. Jr. & J. B. Wallace, 1981. A model of seston capture by net-spinning caddisflies. Oikos 36: 147–157.

Haefner, J. D. & J. B. Wallace, 1981. Production and potential seston utilization by Parapsyche cardis and Diplectrona modesta (Trichoptera: Hydropsychidae) in two streams draining contrasting southern Appalachian watersheds. Envir. Ent. 10: 433–441.

Hamilton, A. L., 1969. On estimating annual production. Limnol. Oceanogr. 14: 771–782.

Hopkins, C. L., 1976. Estimate of biological production in some stream invertebrates. N. Z. J. mar. Freshwat. Res. 10: 629–640.

Hynes, H. B. N., 1980. A name change in the secondary production business. Limnol. Oceanogr. 25: 778.

Hynes, H. B. N. & M. J. Coleman, 1968. A simple method of assessing the annual production of stream benthos. Limnol. Oceanogr. 13: 569–573.

Krueger, C. C. & F. B. Martin, 1980. Computation of confidence intervals for the size-frequency (Hynes) method of estimating secondary production. Limnol. Oceanogr. 25: 773–777.

Kurtak, D. C., 1979. Food of blackfly larvae (Diptera: Simuliidae): seasonal changes in gut contents and suspended material at several sites in a single watershed. Quaest. Ent. 15: 357–374.

Ladle, M., J. A. B. Bass & W. R. Jenkins, 1972. Studies on production and food consumption by the larval Simuliidae (Diptera) of a chalk stream. Hydrobiologia 39: 429–448.

MacFarlane, M. B. & T. F. Waters, 1982. Annual production by caddisflies and mayflies in a western Minnesota plains stream. Can. J. Fish. aquat. Sci. 39: 1628–35.

Maciolek, J. A. & M. G. Tunzi, 1968. Microseston dynamics in a simple Sierra Nevada lake-stream system. Ecology 49: 60–75.

Mackay, R. J., 1978. Larval identification and instar association in some species of Hydropsyche and Cheumatopsyche (Trichoptera, Hydropsychidae). Ann. ent. Soc. Am. 71: 499–509.

Mackay, R. J., 1979. Life history patterns of some species of Hydropsyche (Trichoptera: Hydropsychidae) in southern Ontario. Can. J. Zool. 57: 963–975.

McCullough, D. A., G. W. Minshall & C. E. Cushing, 1979. Bioenergetics of lotic filter-feeding insects Simulium spp. (Diptera) and Hydropsyche occidentalis (Trichoptera) and their function in controlling organic transport in streams. Ecology 60: 585–596.

Merritt, R. W., D. H. Ross & G. J. Larson, 1982. Influence of stream temperature and seston on the growth and production of overwintering larval black flies (Diptera: Simuliidae). Ecology 63: 1322–1331.

Nelson, D. J. & D. C. Scott, 1962. Role of detritus in the productivity of a rock-outcrop community in a piedmont stream. Limnol. Oceanogr. 7: 396–413.

Neves, R. J., 1979. Secondary production of epilithic fauna in a woodland stream. Am. Midl. Nat. 102: 209–224.

Oswood, M. W., 1979. Abundance patterns of filter-feeding caddisflies and seston in a Montana (U.S.A.) lake outlet. Hydrobiologia 63: 177–183.

Parker, C. R. & J. R. Voshell Jr., 1982. Life histories of some filter-feeding Trichoptera in Virginia. Can. J. Zool. 60: 1732–1742.

Parker, C. R. & J. R. Voshell Jr., 1983. Production of filter-feeding Trichoptera in an impounded and a free-flowing river. Can. J. Zool. 61: 70–87.

Rhame, R. E. & K. W. Stewart, 1976. Life cycles and food habits of three Hydropsychidae (Trichoptera) species in the Brazos River, Texas. Trans. am. ent. Soc. 102: 65–99.

Ross, D. H. & J. B. Wallace, 1983. Longitudinal patterns of production, food consumption, and seston utilization by net-spinning caddisflies (Trichoptera) in a southern Appalachian stream (U.S.A.). Holarct. Ecol. 6: 270–284.

Schuster, G. A. & D. A. Etnier, 1978. A manual for the identification of the larvae of the caddisfly genera Hydropsyche Pictet and Symphitopsyche Ulmer in Eastern and Central North America (Trichoptera: Hydropsychidae). U.S.E.P.A., Envir. monit. Ser. EPA-600/4-78-060, 129 pp.

Sheldon, A. L. & M. W. Oswood, 1977. Blackfly (Diptera: Simuliidae) abundance in a lake-outlet: test of a predictive model. Hydrobiologia 56: 113–120.

Stone, A. & E. L. Snoddy, 1969. The black flies of Alabama. Agric. exp. Stat., Auburn Univ., Ala. Bull. 390, 93 pp.

Ulfstrand, S., 1968. Benthic animal communities in Lapland streams. Oikos, Suppl. 10: 1–120.

U.S.G.S., 1980. Water resources data for Georgia. U.S. Geol. Surv. Wat. Data Rep. GA-80-1, 455 pp.

Vannote, R. L. & B. W. Sweeney, 1980. Geographic analysis of thermal equilibria: A conceptual model for evaluating the effect of natural and modified thermal regimes on aquatic insect communities. Am. Nat. 115: 667–695.

Wallace, J. B. & R. W. Merritt, 1980. Filter-feeding ecology of aquatic insects. Ann. Rev. Ent. 25: 103–132.

Wallace, J. B., J. R. Webster & W. R. Woodall, 1977. The role of filter feeders in flowing waters. Arch. Hydrobiol. 4: 506–532.

Ward, J. V. & J. A. Stanford, 1982. Thermal responses in the evolutionary ecology of aquatic insects. Ann. Rev. Ent. 27: 97–117.

Waters, T. F., 1977. Secondary production in inland waters. Adv. ecol. Res. 10: 91–164.

Waters, T. F. & G. W. Crawford, 1973. Annual production of a stream mayfly population: a comparison of methods. Limnol. Oceanogr. 18: 286–296.

Waters, T. F. & J. C. Hokenstrom, 1980. Annual production and drift of the stream amphipod Gammarus pseudolimnaeus in Valley Creek, Minnesota. Limnol. Oceanogr. 25: 700–710.

Williams, N. E. & H. B. N. Hynes, 1973. Microdistribution and feeding of the net-spinning caddisflies (Trichoptera) of a Canadian stream. Oikos 24: 73–84.

Wilm, H. G. & H. C. Storey, 1944. Velocity-head rod calibrated for measuring stream flow. Civ. Engng. 14: 475–476.

Wotton, R. S., 1978. Life-histories and production of blackflies (Diptera: Simuliidae) in moorland streams in Upper Teesdale, Northern England. Arch. Hydrobiol. 83: 232–250.

Wotton, R. S., 1979. The influence of a lake on the distribution of blackfly species (Diptera: Simuliidae) along a river. Oikos 32: 368–372.

Received 15 April 1983; in revised form 26 August 1983; accepted 20 September 1983.